

Indices and surds - Set 1 - Solutions

Specification MM1.1, MM1.2, M2.7, M2.11, M2.12.

Answer key

Question	Answer	Topic
1	F	Indices and surds
2	C	Indices and surds
3	A	Indices and surds
4	B	Indices and surds
5	E	Indices and surds
6	B	Indices and surds
7	E	Indices and surds
8	E	Indices and surds
9	A	Indices and surds
10	C	Indices and surds
11	E	Indices and surds
12	B	Indices and surds

Worked solutions

Question 1 IND-01 · Indices and surds · spec MM1.1

Simplify $\frac{(2x^3)^4 \times 8x^{-5}}{4x^2}$.

Answer: F

$(2x^3)^4 = 16x^{12}$, so the numerator is $16x^{12} \times 8x^{-5} = 128x^7$. Dividing by $4x^2$ gives $32x^5$.

Question 2 IND-02 · Indices and surds · spec MM1.1

Find the value of $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$.

Answer: C

The negative index inverts the fraction: $\left(\frac{8}{27}\right)^{\frac{2}{3}}$. Cube root first: $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$.

Question 3 IND-03 · Indices and surds · spec MM1.1

Solve $4^{x+1} = 8^{2x-3}$.

Answer: A

Write both sides as powers of 2: $2^{2x+2} = 2^{6x-9}$. So $2x+2 = 6x-9$, giving $4x = 11$ and $x = \frac{11}{4}$.

Question 4 IND-04 · Indices and surds · spec MM1.2

Express $\frac{5}{3+2\sqrt{5}}$ in the form $a+b\sqrt{5}$ where a and b are rational.

Answer: B

Multiply top and bottom by $3-2\sqrt{5}$: denominator $9-20 = -11$, numerator $15-10\sqrt{5}$. So the value is $\frac{15-10\sqrt{5}}{-11} = \frac{-15+10\sqrt{5}}{11}$.

Question 5 IND-05 · Indices and surds · spec MM1.2

Simplify $\sqrt{75} - \sqrt{27} + \sqrt{12}$.

Answer: E

$\sqrt{75} = 5\sqrt{3}$, $\sqrt{27} = 3\sqrt{3}$, $\sqrt{12} = 2\sqrt{3}$, so the total is $(5-3+2)\sqrt{3} = 4\sqrt{3}$.

Question 6 IND-08 · Indices and surds · spec MM1.1

Solve $9^x - 4 \times 3^x + 3 = 0$.

Answer: B

Let $y = 3^x$: $y^2 - 4y + 3 = (y-1)(y-3) = 0$, so $3^x = 1$ or $3^x = 3$, giving $x = 0$ or $x = 1$.

Question 7 IND-10 · Indices and surds · spec MM1.1

The value of $\frac{2^{n+2} + 2^n}{2^{n+1}}$ is

Answer: E

Factor 2^n : numerator $= 2^n(4+1) = 5 \cdot 2^n$; denominator $= 2 \cdot 2^n$. Quotient $= \frac{5}{2}$.

Question 8 IND-13 · Indices and surds · spec MM1.1Simplify $\left(\frac{a^{\frac{1}{2}}b^{-2}}{a^{-1}b^{\frac{1}{2}}}\right)^4$.**Answer: E**Inside the bracket: $a^{\frac{1}{2}+1}b^{-2-\frac{1}{2}} = a^{\frac{3}{2}}b^{-\frac{5}{2}}$. Raise to the 4th power: a^6b^{-10} .**Question 9** IND-14 · Indices and surds · spec MM1.2The number $\frac{\sqrt{50} + \sqrt{18}}{\sqrt{8}}$ is equal to**Answer: A** $\sqrt{50} = 5\sqrt{2}$, $\sqrt{18} = 3\sqrt{2}$, $\sqrt{8} = 2\sqrt{2}$. So $\frac{8\sqrt{2}}{2\sqrt{2}} = 4$.**Question 10** IND-15 · Indices and surds · spec MM1.1Given that $2^x = 5$ and $2^y = 20$, find the value of 2^{2x-y} .**Answer: C**

$$2^{2x-y} = \frac{(2^x)^2}{2^y} = \frac{25}{20} = \frac{5}{4}.$$

Question 11 IND-17 · Indices and surds · spec MM1.1Solve $x^{\frac{3}{2}} = 8x^{-\frac{3}{2}}$ for $x > 0$.**Answer: E**Multiply by $x^{3/2}$: $x^3 = 8$, so $x = 2$.**Question 12** IND-19 · Indices and surds · spec MM1.1How many real solutions does the equation $2^{x^2} = 16 \times 2^{-3x}$ have?**Answer: B** $2^{x^2} = 2^{4-3x}$, so $x^2 + 3x - 4 = 0$, i.e. $(x+4)(x-1) = 0$: two real solutions.